ETH zürich

Mobile Computing

Introduction - Wireless Standards

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Course Outline

Introduction (WIRELESS STANDARDS)

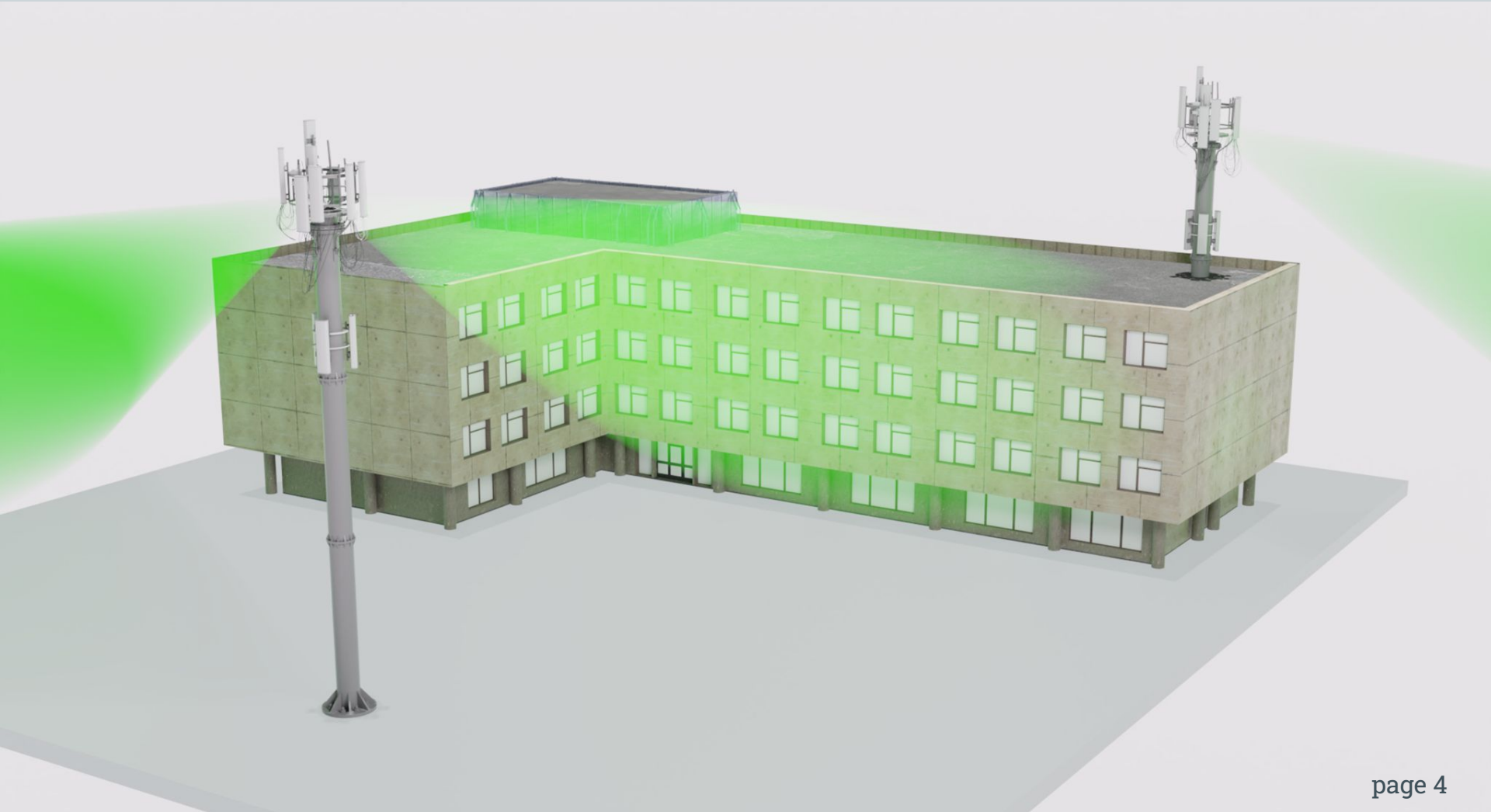
1. Data and Signals
2. Data Communication
3. Mobile Systems
4. Mobile Algorithms
5. Mobile AI
6. Cloud Computing

Tournament

Intro - Focus in this course: Wi-Fi

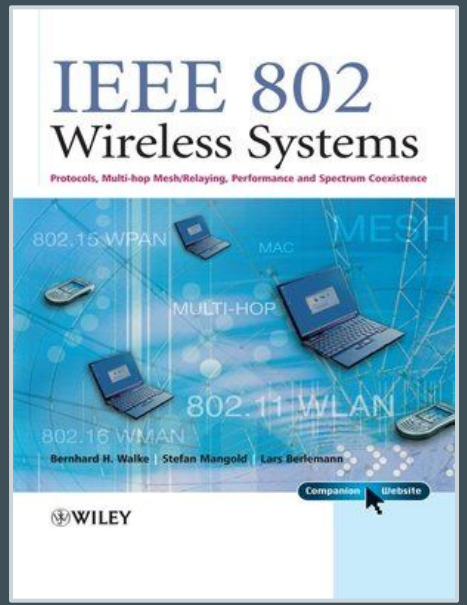


Intro - Also discussed in this course: 3GPP



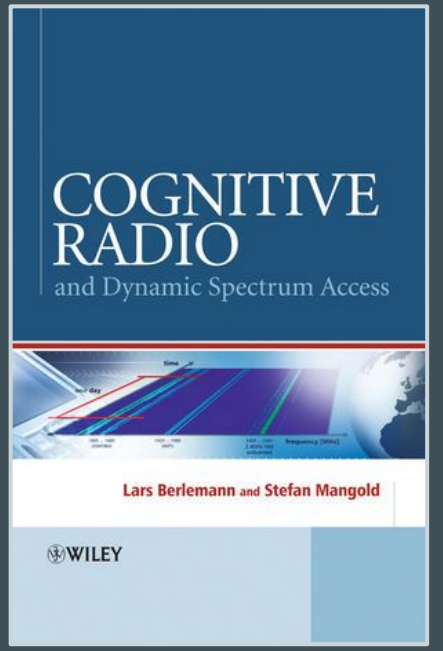
Intro - Textbook (1/2)

- Bernhard Walke, Stefan Mangold, Lars Berlemann (2006)
  IEEE 802 Wireless Systems – Protocols, Multi-Hop Mesh / Relaying, Performance and Spectrum Coexistence
- November 2006, 402 pages.
  John Wiley & Sons
- Available at ETH library



Intro - Textbook (2/2)

- Lars Berlemann, Stefan Mangold (2009)
  Cognitive Radio and Dynamic Spectrum Access
- January 2009, 256 pages.
  John Wiley & Sons
- Available at ETH library

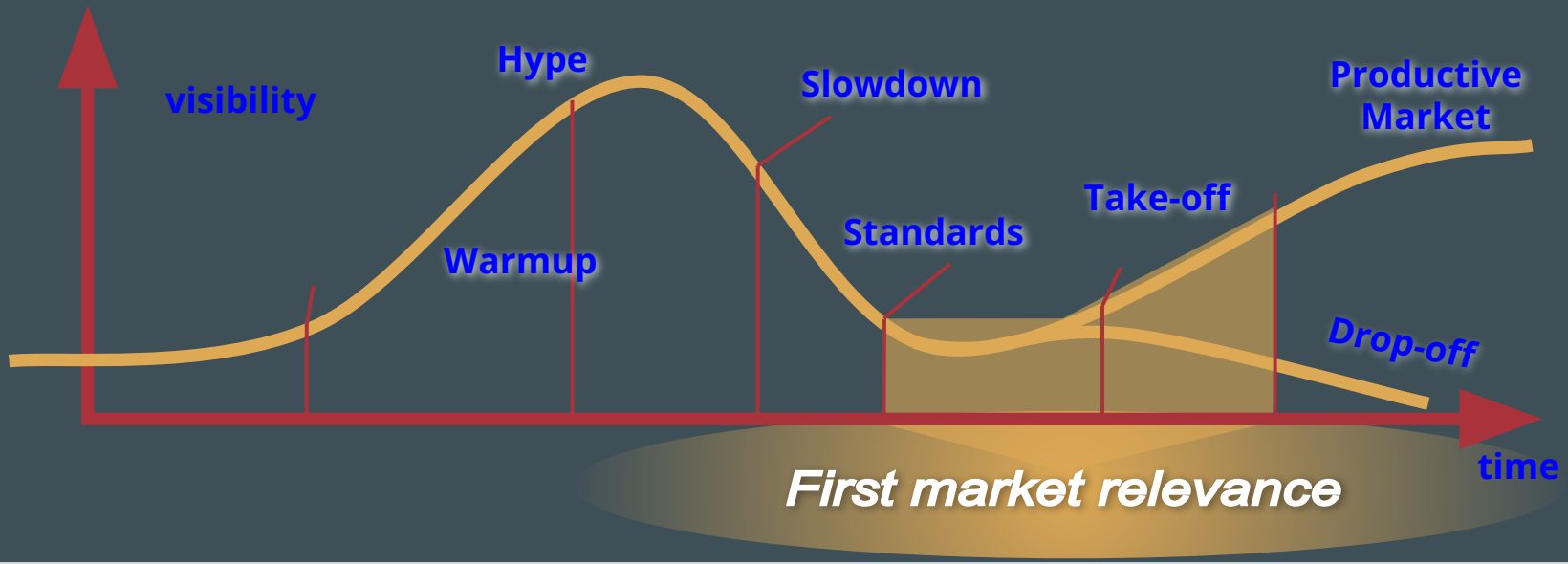


Simulator

- System emulation software (Java), available from Bitbucket: Jemula802++ and Jemula++
- Getting started
  - lfield.ch/wiki/lovefield_jemula802.html
  - bitbucket.org/lfield/jemula802/wiki/Home
- Helpful tools: Jupyter (“Wireless” Notebooks) with Python examples

Life-Cycle of Technology

- In this course we will look at technology that ...
  - ... took off into productive markets → 5G, Wi-Fi 6, Bluetooth 5
  - ... or, is at its hype → Satellite, Wi-Fi 7+, Mobile A.I.
  - ... or, is warming up → Li-Fi, Mobile A.I.



Terminology

- 1 Mb/s = 1 Megabit per second
  - not: Mbit/s or Mbits/s
  - Mbps <- recommended only for source code
- 1 MB/s = 1 Megabyte per second
  - not: MByte/s or MBytes/s
  - MBps only for source code
- same for kilo, Giga, ...
  - kilobyte = kB kilobit = kb Gigabit = Gb
- 1 second = 1 s
  - not: sec ... secs

From Research to Commercialization

- Academic and industry research
- Standardization (IEEE, 3GPP, ETSI...)
- Certification (Wi-Fi Alliance, Bluetooth Forum, 5G ...)
- Spectrum regulation (FCC, BAKOM, ERO, ITU...)
- Manufacturing and development (semiconductor- and device industry)
- Deployment (telecom operators, consumer markets)

5 ... 10 Years Process

Standardization

- Commercial and industrial players develop a new technology together
  - to achieve interoperability, which increases the product value
  - shortens time-to-market
- Relevant standardization bodies
  - Institute of Electrical and Electronics Engineers (IEEE)
  - 3rd Generation Partnership Project (3GPP)
  - European Telecommunications Standards Institute (ETSI)
- International Telecommunication Union (ITU)
  - Defines mobile standards, helps global harmonizing

Wi-Fi and 3GPP Standards

("G"=Generation)

Wi-Fi: Wireless LAN

- 1999: 802.11b products (11 Mb/s)
- 2000: Wi-Fi certification
- 2001: 802.11a products (54 Mb/s)
- 2005: 802.11e QoS
- 2010: 802.11af TV White Space
- 2010: 802.11ad 60GHz
- 2016: 802.11ah IoT
- 2017: 80211ax Multi-User MIMO
- 2024: 802.11be Wi-Fi 7 approved
- 2025: 802.11bf sensing
- 2028: 802.11bn Wi-Fi 8 high reliability

3GPP: Cellular networks

- 1980s: analog cellular
- 1990s: 2G digital cellular, GSM
- 2005: 3G: UMTS 384 kb/s
- 2010: 4G LTE: End-to-end IP core, Unlicensed LTE (LTE-U)
- 2015: Internet-of-Things, LoRa
- 2020: 5G
- 2022: Beyond 5G, edge comp., AI
- 2023-25: Pre-6G: intelligent surfaces
- 2024: 5G-Advanced (Release 18)
- 2025–2027: 6G studies (Release 20)
- 2030: 6G / IMT-2030 (target)

ITU-IMT Wireless System Framework

- ITU = “International Telecommunications Unit”
- IMT = standard family “International Mobile Telecommunication”
- IMT-2000 (3G UMTS / EDGE / DECT)
  - deployment started in 2000
- IMT-Advanced
  - 1 Gb/s @low mobility
  - 4G LTE and 802.11 standards
- IMT-2020
  - 3GPP 5th Generation (5G)



Institute of Electrical and Electronics Engineers

- Non-profit organization, for various industrial sectors
  - Aerospace
  - Biomedical
  - Consumer electronics
  - Computers, Telecommunication
  - Electrical power
- Academic publication platform
  - Publishes journals and magazines
  - Organizes of conferences
- Similar to the Association for Computing Machinery, ACM



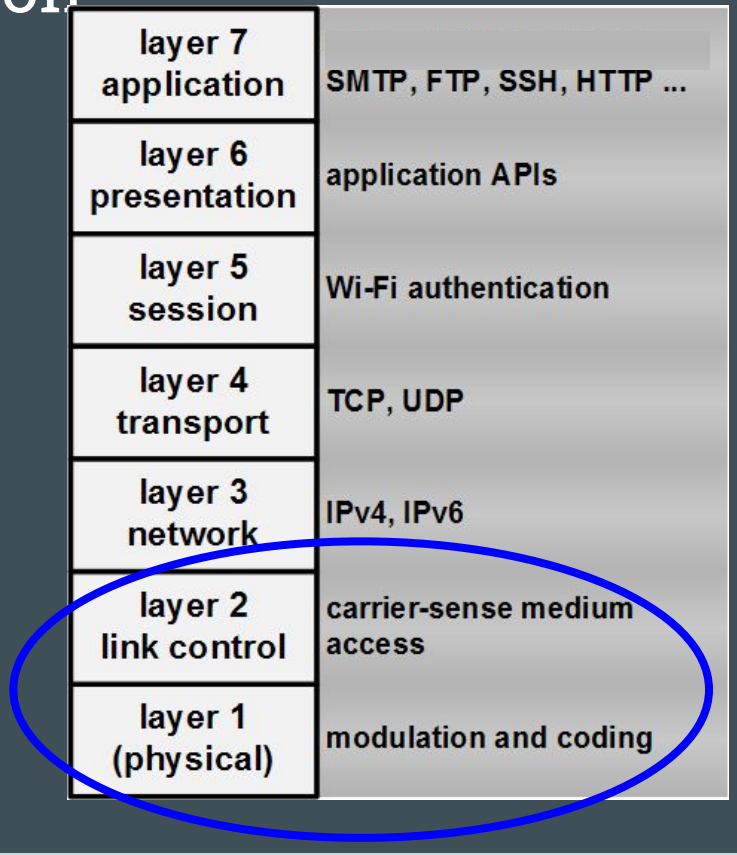


The Standardization Project IEEE 802

- LAN/MAN Standards Committee
- Focus is mainly on lower two of the ISO/OSI layers
  - Layer 1 (physical) and layer 2 (link control)
- Organized in Working Groups
  - 802.1 Higher Layer Local Area Network
  - 802.3 Ethernet
  - 802.11 Wireless Local Area Networks → Wi-Fi
  - 802.15 Wireless Personal Area Networks → Bluetooth, Ultra Wideband

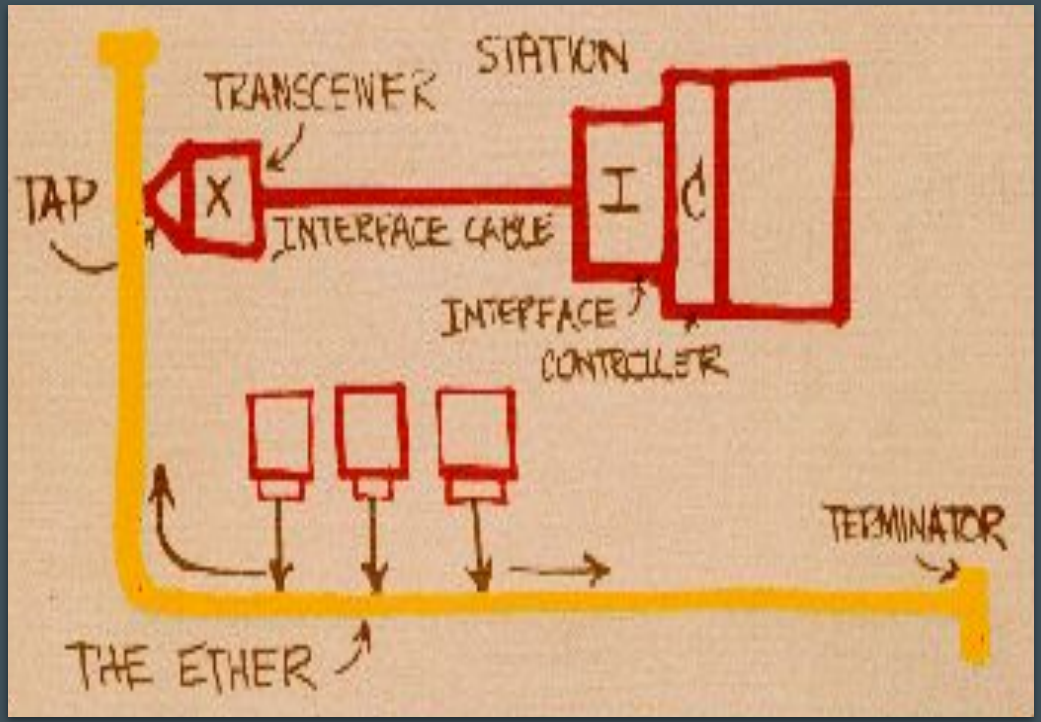
ISO/OSI Reference Model - Layer 1 and Layer 2

- ISO: International Organization for Standardization
- OSI: Open Systems Interconnection
- Wi-Fi IEEE 802.11 is for layer 1 and layer 2



Ethernet

Early drawing from ~1976 (Bob Metcalfe)

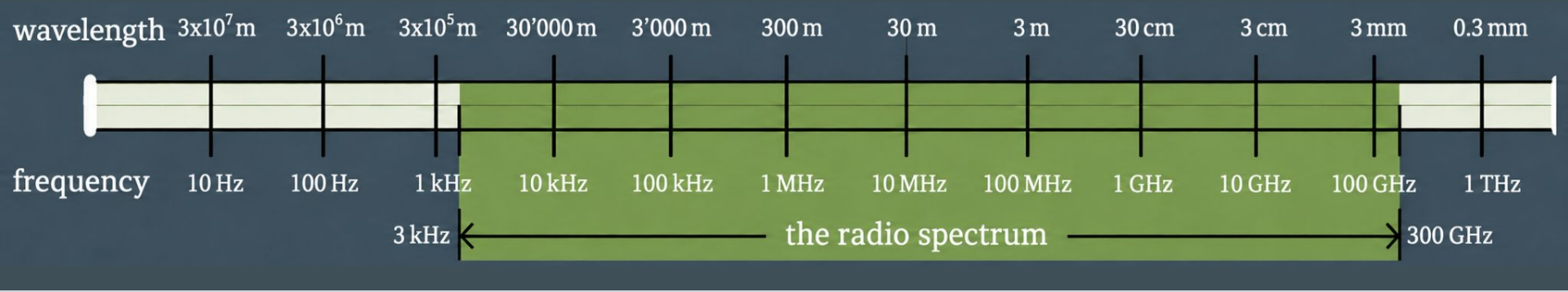


Cellular Standards, 3GPP Releases

2018	Rel-15	5G Basic: eMBB (enhanced Mobile Broadband) URLLC (Ultra Reliable Low Latency Communications)
2019	Rel-16	5G Evolution: V2X (Vehicle to X), New Radio Unlicensed (NR-U), Industrial IoT (IIoT), Time-Sensitive Networking, Integrated Access and Backhaul (IAB), Positioning
2021	Rel-17	5G Evolution: mMTC (massive Machine Type Communications)
2023	Rel-18	5G Advanced: Reduced Capability (RedCap), Non-Terrestrial-Networks (NTN)
2025	Rel-19	5G Advanced
2026	Rel-20	5G Advanced
2028	Rel-21	6G Basic: Terahertz Communication

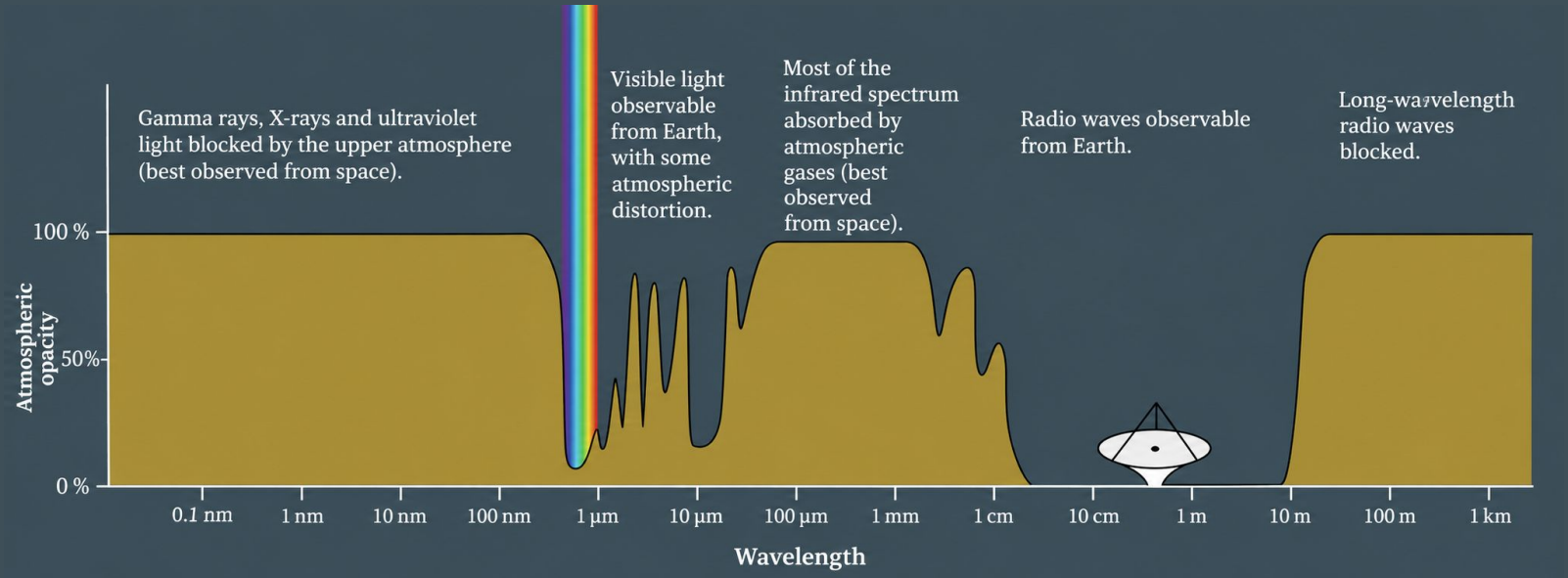
Radio Frequency Spectrum

- Electromagnetic frequencies between 3 kHz and 300 GHz are commonly referred to as “radio” frequencies
- In vacuum, frequency (f) and wavelength (ƛ) are inversely proportional to each other. A wave with higher frequency has a shorter wavelength: **ƛ = c / f**

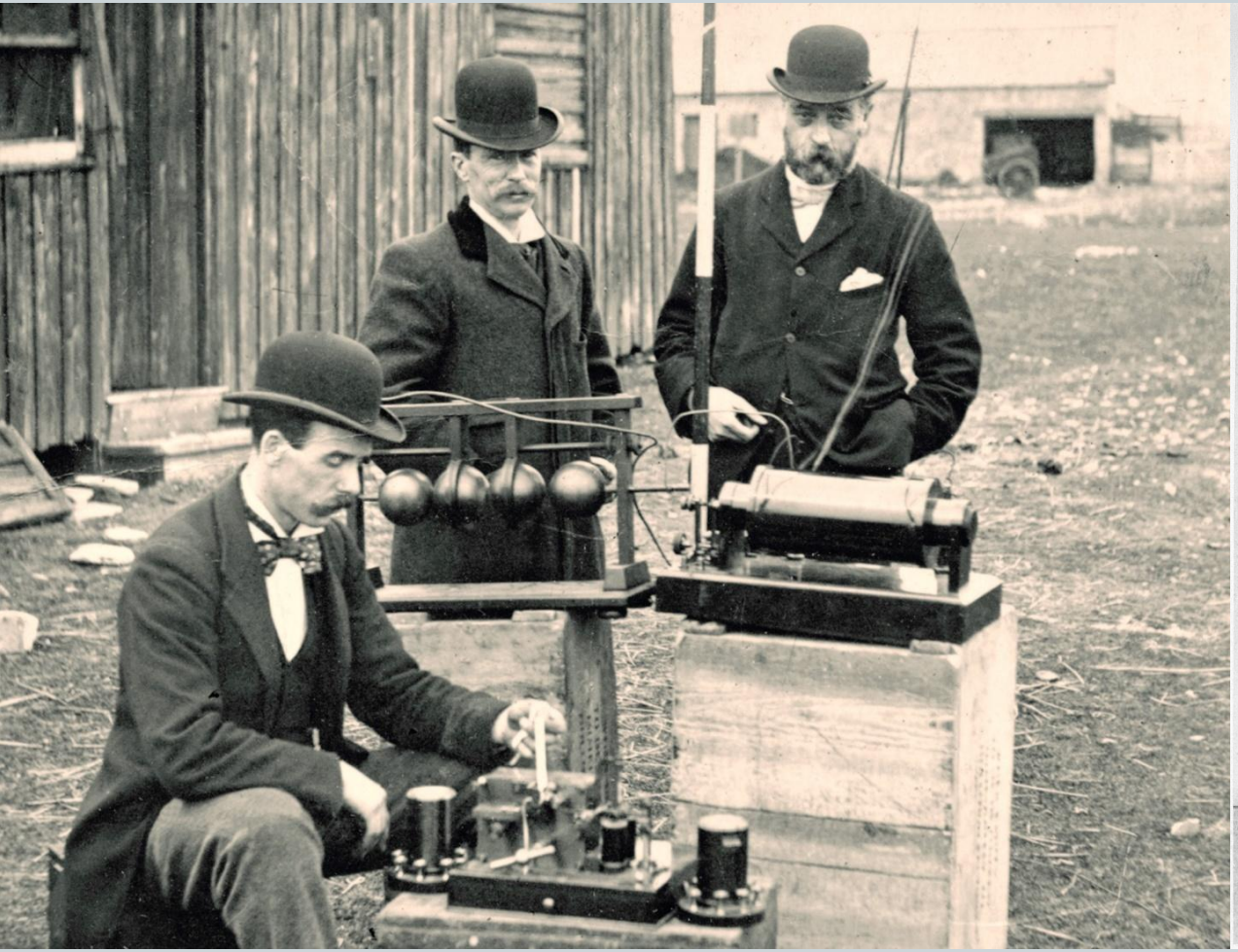


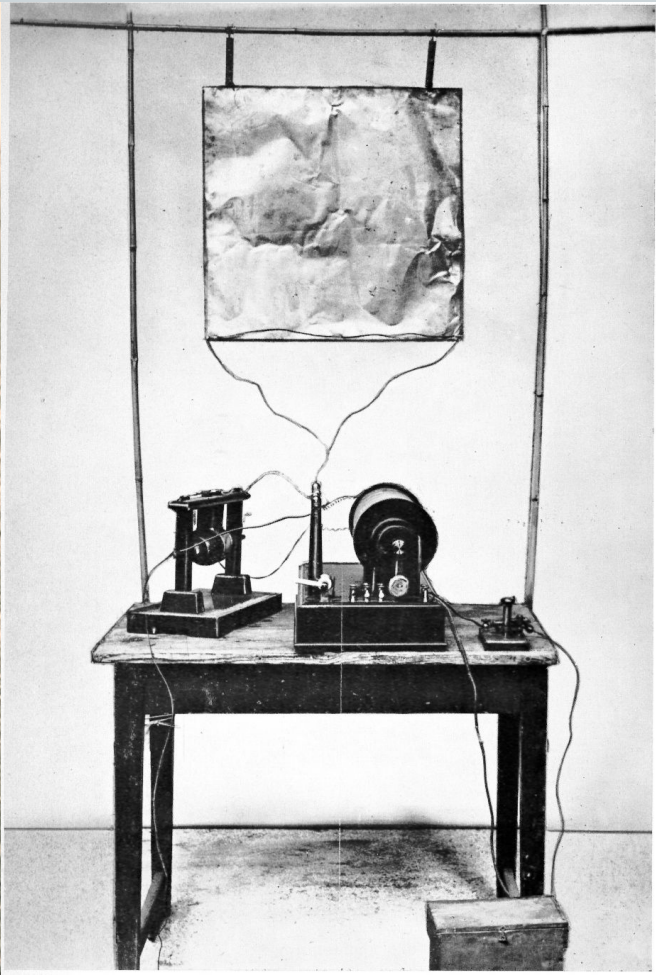
Atmospheric Opacity for Electromagnetic Waves

- The Earth's atmosphere blocks or absorbs electromagnetic waves, due to factors like water vapor, carbon dioxide, ozone, and wavelength

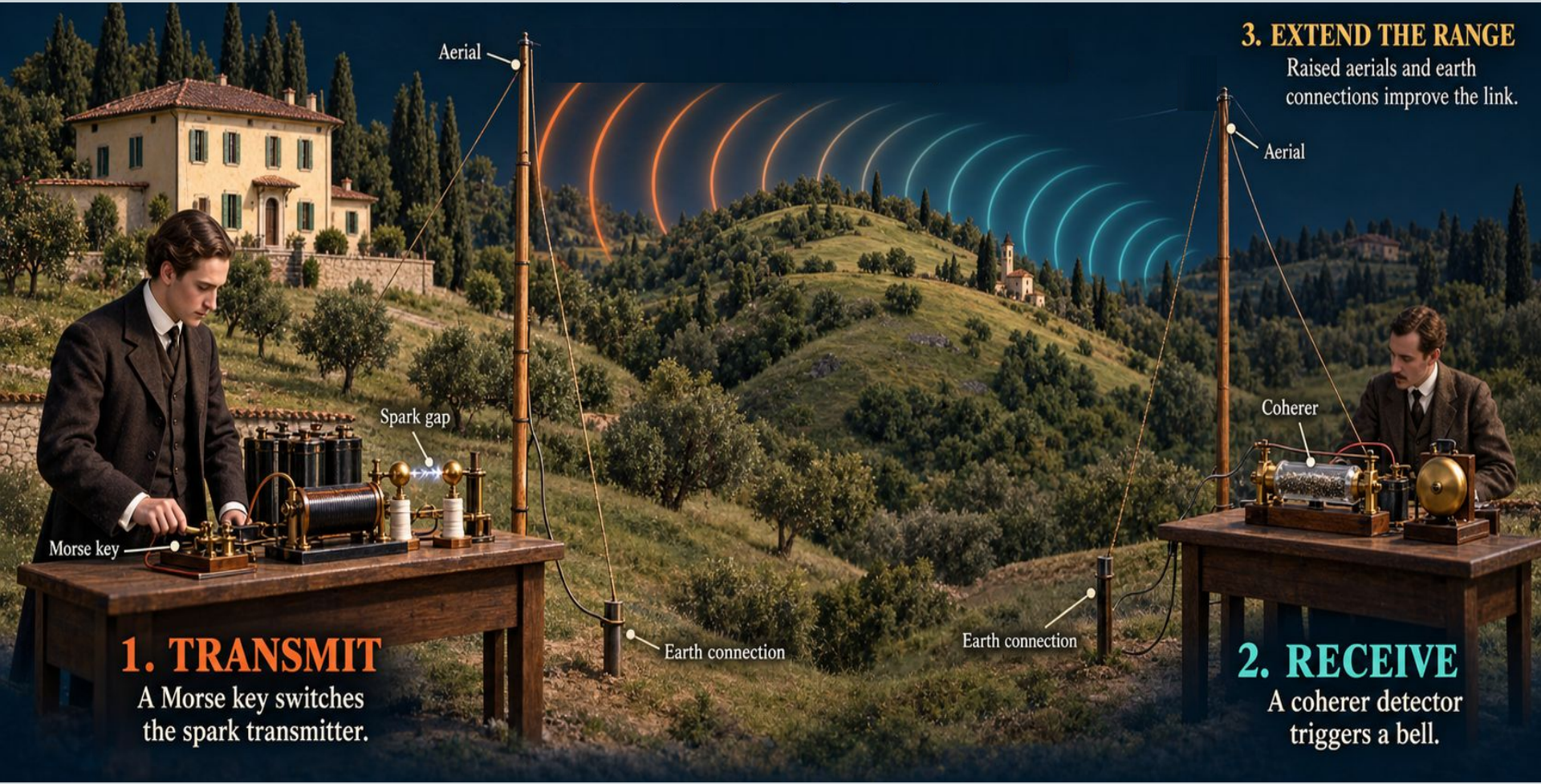


Guglielmo Marconi 1895, Bologna





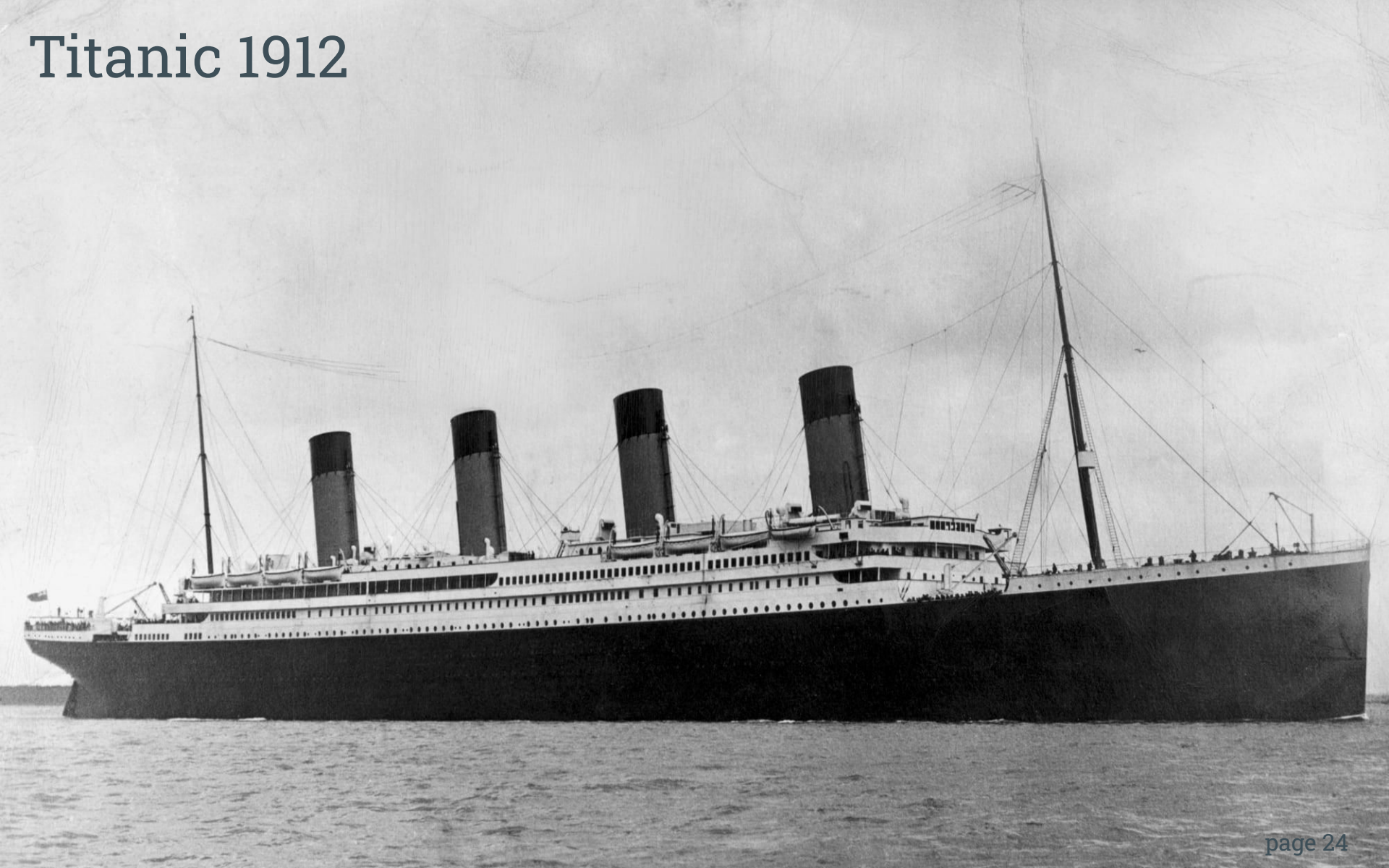
Marconi: Sending Wireless Signals



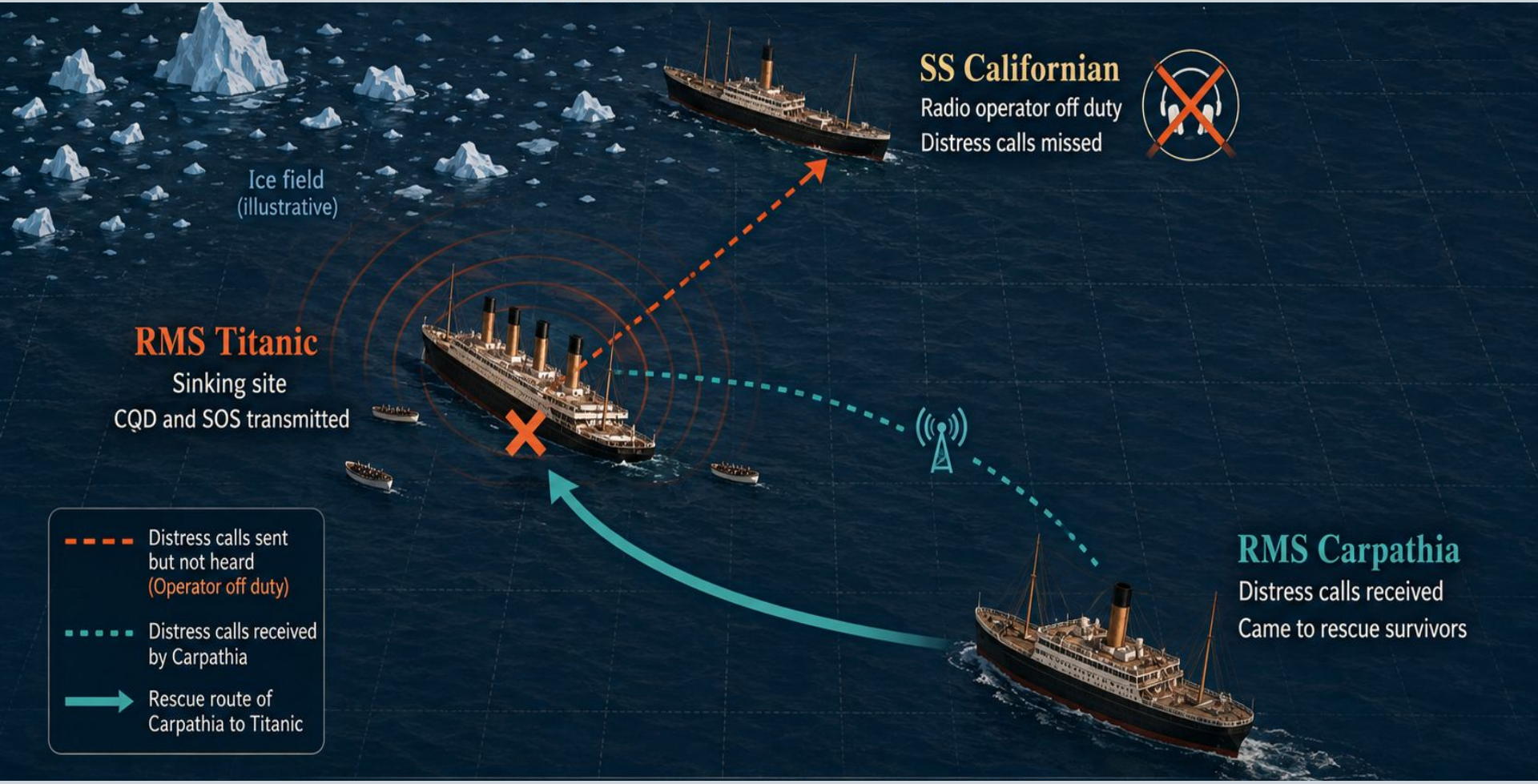
Guglielmo Marconi 1895, Bologna

- Inventor and electrical engineer, from Italy
- Developed the **first practical system of wireless telegraphy**, enabling long-distance communication without the need for physical wires
- First demonstrated in Bologna, 1895
- His work earned him the Nobel Prize in Physics in 1909, which he shared with Karl Ferdinand Braun

Titanic 1912



Missed Distress Calls



Titanic (1912): Radio and Regulation

14-15 April 1912: distress calls and missed radio watch

- Titanic sent SOS using Morse radiotelegraphy
  - Carpathia received the calls, diverted, and rescued about 700 survivors after Titanic sank
  - Nearby Californian missed the calls: its sole operator had gone off duty and was no longer listening
- 1912 reforms strengthened radio licensing, distress priority, and shipboard watchkeeping
- SOLAS 1914 included continuous radio watches
  - first International Convention for the Safety of Life at Sea

Regulatory Bodies (Government Agencies)

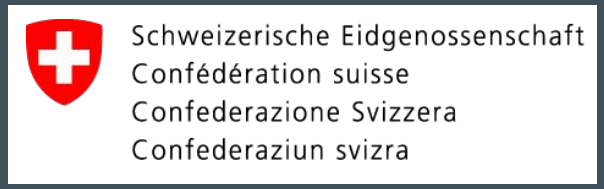
- International Telecommunications Union (ITU)
- United States
  - National Telecom. & Information Administration (NTIA)
  - Federal Communications Commission (FCC)
- European Conference of Postal and Telecommunications Administrations
- Swiss Federal Office of Comm., OFCOM/BAKOM





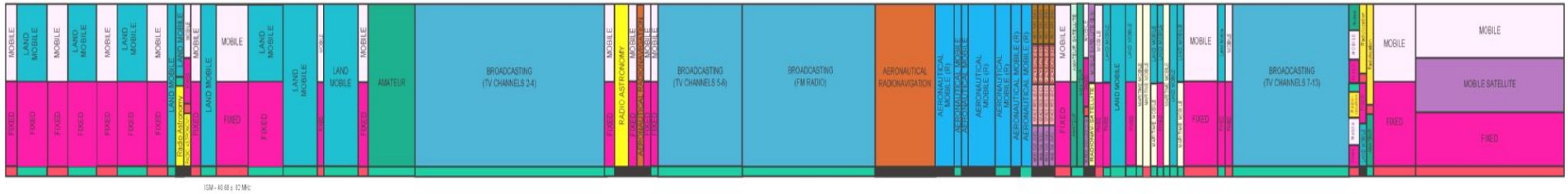




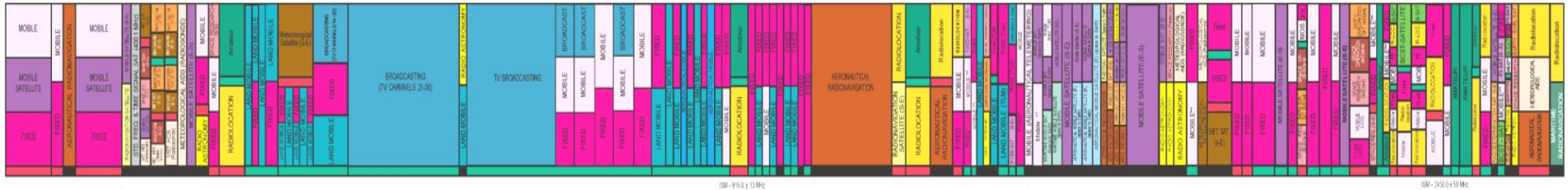


Radio Spectrum Allocation

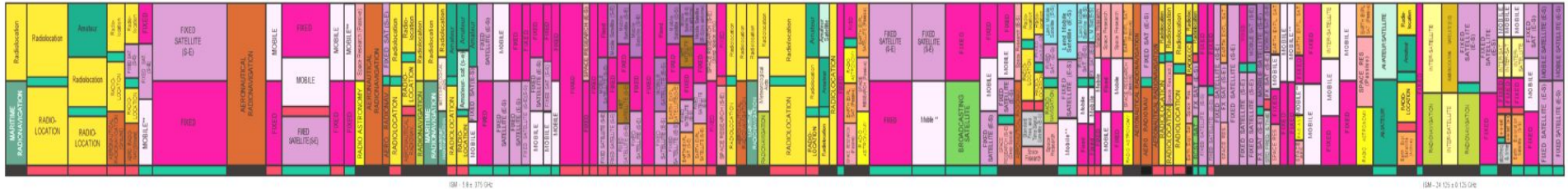
30 MHz … → **→ … 0.3 GHz**



0.3 GHz … → **→ … 3 GHz**



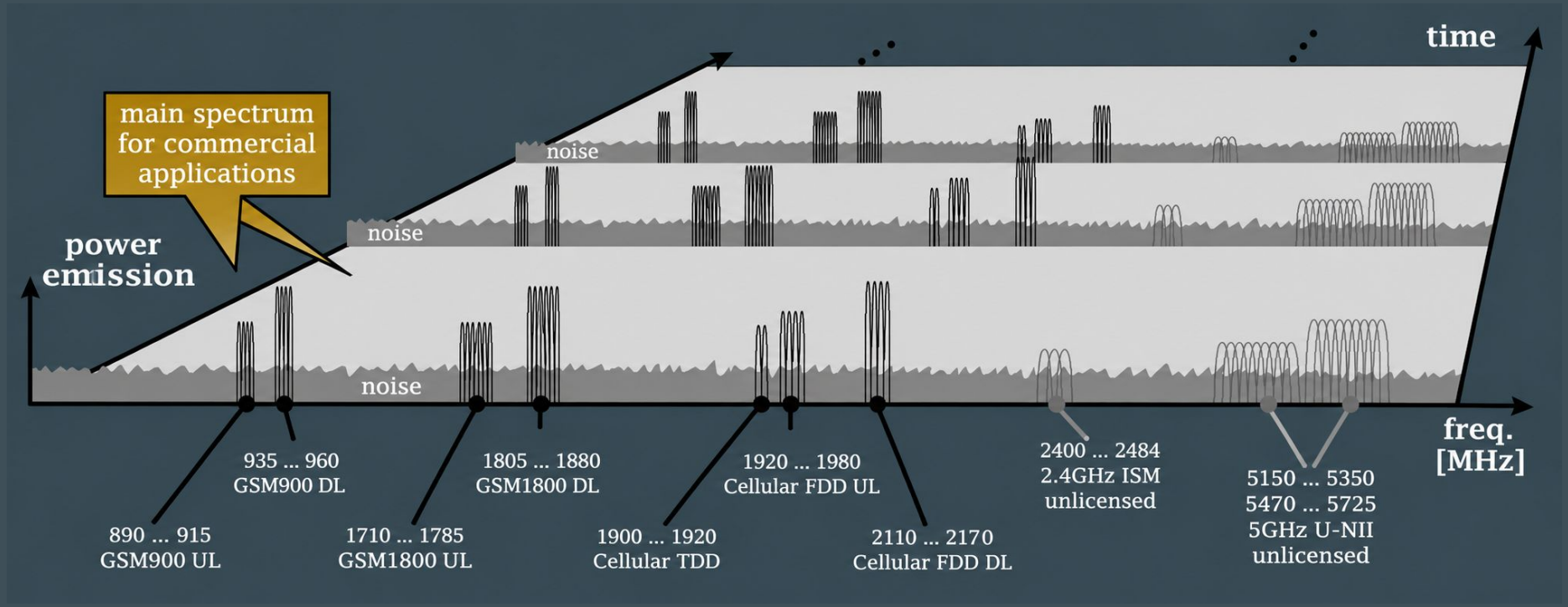
3 GHz … → **→ … 30 GHz**



Licensed Radio Spectrum

- Most part of the radio spectrum is licensed
  - Exclusive access to spectrum, commercial license holders pay fees
  - Considered interference-free, reliable communication
- Licenses are granted for long times (10y +)
  - Exclusive usage rights might lead to unused spectrum or inefficient spectrum use

Licensed Cellular Radio Spectrum (now 5G)



Unlicensed Radio Spectrum

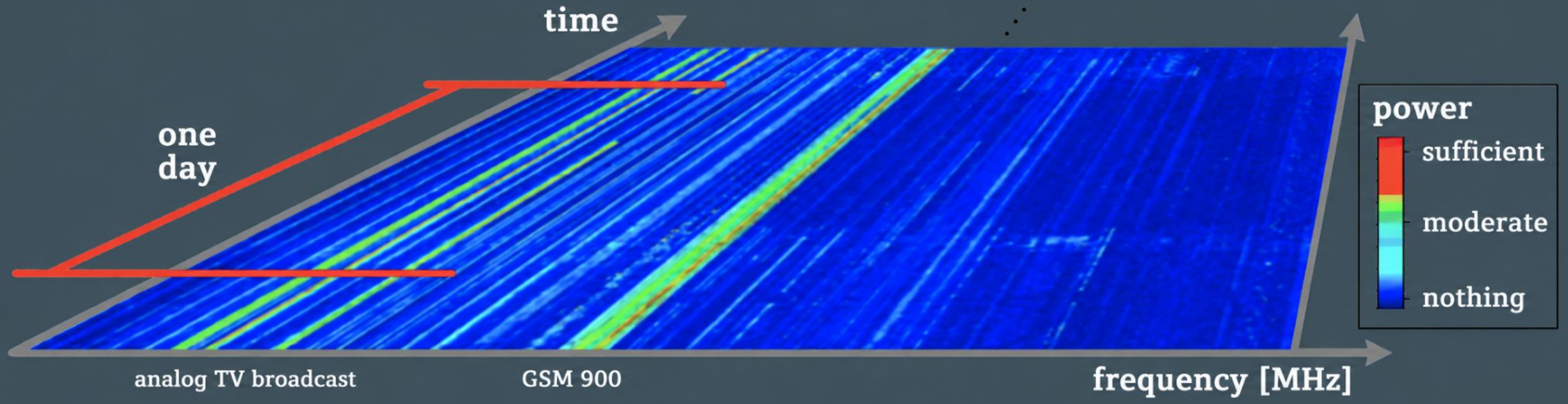
- Open access, but controlled access
  - Systems need to comply with technical rules of spectrum access, or standards, to keep the potential radio interference limited.
  - Radio regulation does not impose requirements about how to access spectrum, and what technology to use; instead stakeholders (users, industry) agree on common standards.
- Limitations
  - The unlicensed spectrum can be overcrowded, neighbors might interfere with each other.
  - The unlicensed spectrum is a shared resource. The quality of communication cannot be guaranteed by the system.

Unlicensed Radio Spectrum – FCC Part 15 Rules

- Three principles
  - (i) listen before talk
  - (ii) when talking, make frequent pauses and listen again
  - (iii) do not talk too loud
- When detecting a used channel, another unused channel should be used
  - or the radio device has to wait until the channel has become idle again
- Rules do not require information exchange between devices

Secondary Radio Spectrum Use

- Ultra Wideband (UWB), below noise
- White Space communication
  - above noise, opportunistic usage
- Cognitive Radio



Mobile Computing Algorithms

- Standards define common technical specifications and may incorporate patented innovations
  - Developed among manufacturers, users, and other stakeholders
  - Often shaped by stakeholders' commercial interests
  - Dominant market players may favour proprietary solutions when interoperability offers little benefit
- Regulation defines mandatory requirements
  - Special exemptions apply in specific contexts, such as defence
- Algorithms determine how systems operate
  - Enable competing implementations
  - Can be developed without industry-wide consensus

Exercise: Scientific Literature

- Work through notebook 00.3
  - Scientific Literature and Research Integrity
  - Check a Titanic claim against credible sources
  - Find and cite two papers on LLMs
  - Compare manual and AI-assisted searches
- Save and upload your notebook to Moodle

